

PHÂN TÍCH THỐNG KÊ NHIỀU CHIỀU (MAT3452)

Bài tập tuần 4: Vector ngẫu nhiên - Mẫu ngẫu nhiên nhiều chiều

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Câu 1. Sinh 500 quan sát từ phân bố chuẩn nhiều chiều với:

(i) Vector kỳ vọng $\mu = (3, 5)^T$

(ii) Ma trận hiệp phương sai $\Sigma = \begin{pmatrix} 2 & 0.5 \\ 0.5 & 1 \end{pmatrix}$

Mô phỏng dữ liệu vừa sinh bằng biểu đồ tán xạ (scatterplot) và biểu đồ tần số / tần suất (histogram) với gói lệnh **ggplot2**. Sau đó tính toán và giải thích hiệp phương sai và tương quan.

```
library(MASS)
library(showtext)

## Loading required package: sysfonts
## Loading required package: showtextdb
library(ggplot2)

font_add("LM", "C:/Windows/Fonts/cambria.ttc")
showtext_auto()
set.seed(100)

mu <- c(3, 5)

sigma <- matrix(c(2, 0.5, 0.5, 1), nrow = 2)

n <- 500

samples <- mvrnorm(n = n, mu = mu, Sigma = sigma)

samples_df <- as.data.frame(samples)
colnames(samples_df) <- c("X1", "X2")

samples_df

##           X1           X2
## 1  3.19644718  6.475319
## 2  2.92709799  4.665378
## 3  2.99152315  5.326833
## 4  1.12472129  6.084694
## 5  2.92219489  4.733738
## 6  2.43904909  5.117286
## 7  4.62750833  3.329444
## 8  2.22103099  4.106676
## 9  3.93636136  5.943202
## 10 3.63248958  4.870073
## 11 2.86120619  4.986127
## 12 3.01809938  4.582553
## 13 4.03847594  3.275670
## 14 1.78479723  5.061589
## 15 3.03489316  4.436783
```

## 16	3.47444463	3.968401
## 17	3.22302648	5.971156
## 18	1.83012641	5.841105
## 19	4.15620688	5.756231
## 20	-0.14322328	3.619500
## 21	3.71643959	4.971092
## 22	2.14420771	4.099869
## 23	2.32210135	5.619620
## 24	2.52687826	3.139746
## 25	4.65900530	4.156347
## 26	3.24827472	6.102742
## 27	4.11445034	5.105485
## 28	2.83249452	4.507833
## 29	4.28985251	6.380497
## 30	2.92016609	4.233551
## 31	3.14696223	4.998918
## 32	1.37114735	2.110007
## 33	2.74633592	6.147862
## 34	2.78472240	5.951396
## 35	4.45423587	4.167901
## 36	3.22578813	5.315937
## 37	2.97558046	4.348879
## 38	2.03181625	5.717292
## 39	1.90623043	3.504545
## 40	1.81290668	4.099430
## 41	3.40393882	4.419345
## 42	0.82774160	4.796853
## 43	5.61217880	5.591347
## 44	2.19777204	4.518687
## 45	3.82223681	5.042526
## 46	1.29303154	3.987891
## 47	3.73998681	4.624443
## 48	1.03194225	4.630504
## 49	3.51862272	3.577977
## 50	5.28481315	6.777192
## 51	3.25519341	6.119470
## 52	5.56742588	5.551179
## 53	2.78083490	4.834732
## 54	0.04641682	4.764339
## 55	6.16744657	6.173057
## 56	1.57899520	4.624303
## 57	4.58691874	6.599286
## 58	0.60952769	3.686687
## 59	1.64501253	2.908824
## 60	4.58406735	4.432270
## 61	4.34348532	2.773646
## 62	3.32731126	4.477063
## 63	3.28934972	5.772330
## 64	-0.59969858	3.666896
## 65	3.06360299	4.342414
## 66	3.61961286	6.272190
## 67	1.79848011	5.423938
## 68	2.40108808	5.662905
## 69	3.17256288	4.854824
## 70	3.24943414	4.756872
## 71	2.15769432	5.290797

## 72	4.41776872	5.709189
## 73	5.15512778	4.309744
## 74	0.56452925	4.479942
## 75	5.65364336	6.598899
## 76	3.44036193	3.887376
## 77	4.52249503	5.546317
## 78	2.81188869	4.403867
## 79	2.12432862	3.199099
## 80	6.14599145	5.458050
## 81	1.93804710	4.082185
## 82	2.98433350	5.231913
## 83	4.59821694	6.364407
## 84	5.94856644	5.378779
## 85	1.76688425	5.222305
## 86	3.35403033	4.758306
## 87	2.52182934	5.314434
## 88	1.87813673	4.535298
## 89	0.90817267	3.344967
## 90	3.23413426	4.837601
## 91	3.63246138	5.635934
## 92	0.92892579	4.455139
## 93	4.79810309	4.125587
## 94	3.60718434	8.027989
## 95	3.93985547	4.789677
## 96	-1.10769267	5.422347
## 97	4.04971696	5.697630
## 98	2.11820454	5.523498
## 99	4.51342874	5.922081
## 100	5.47577187	3.580734
## 101	4.11606129	3.598047
## 102	1.12477528	4.235385
## 103	3.21733567	6.296604
## 104	2.10713270	3.883405
## 105	5.27509650	5.167578
## 106	3.03872205	6.460563
## 107	4.26530409	4.959452
## 108	3.22041554	5.901533
## 109	1.03416134	4.931698
## 110	2.73563359	6.055311
## 111	2.76952199	4.886390
## 112	2.18461755	5.980108
## 113	3.16532914	6.986571
## 114	4.22692161	7.586373
## 115	3.47716979	5.132906
## 116	3.11692310	4.219309
## 117	1.33567075	5.065554
## 118	3.16126338	5.602850
## 119	3.65615120	4.590420
## 120	0.84835437	3.924124
## 121	4.62069208	4.090972
## 122	2.35822129	4.903351
## 123	3.67798012	5.630179
## 124	3.16692228	2.985784
## 125	5.07661978	5.985530
## 126	3.26460135	6.375058
## 127	3.60884491	4.616310

128 1.64850080 4.351710
129 3.79719108 4.898350
130 2.77652375 4.383039
131 4.12573630 3.904172
132 4.61742850 4.396486
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142 5.64259651 6.914392
143 2.32811268 6.013274
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145 4.22618218 5.851360
146 4.54537530 5.592463
147 3.23044047 6.141516
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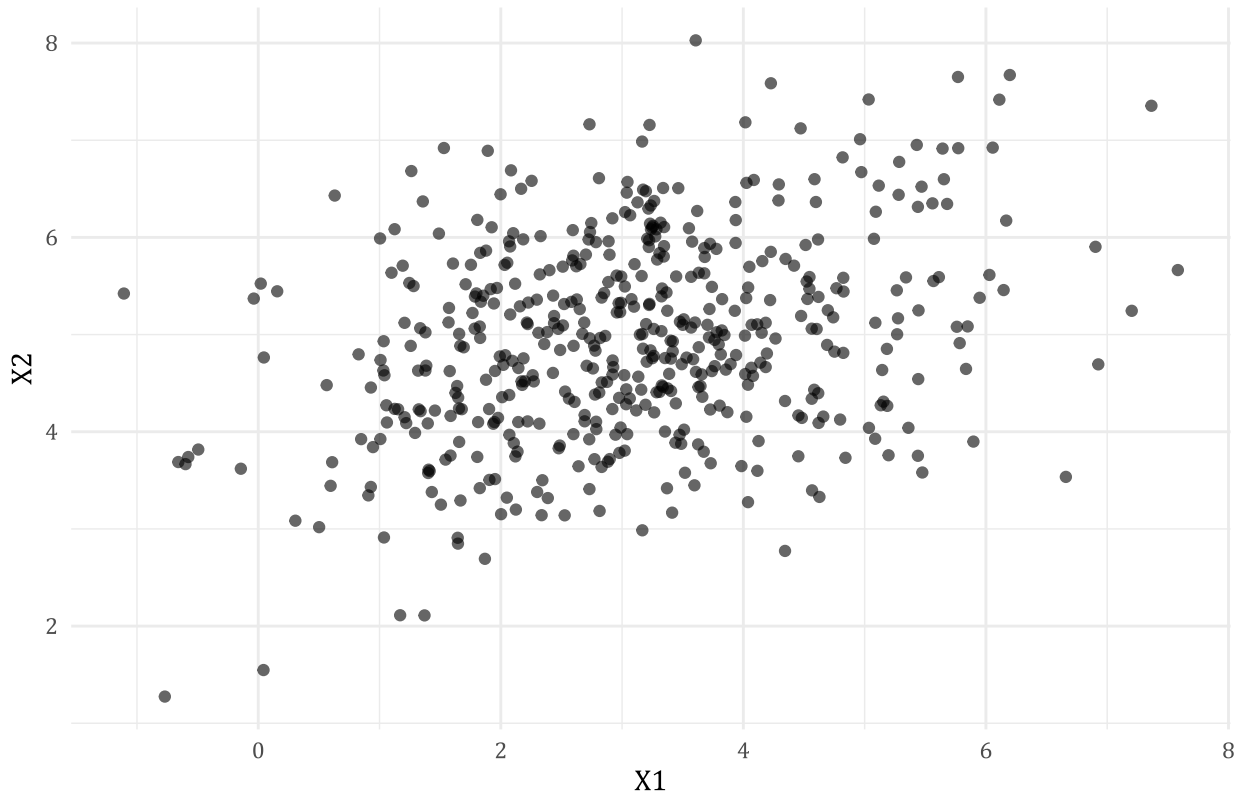
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```
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## 467 4.02466388 4.154433
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## 470 2.88726281 5.540892
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## 472 4.18416360 5.121664
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## 474 5.78395497 4.912200
## 475 4.01534914 4.593566
## 476 0.50226020 3.018241
## 477 2.31094856 5.018542
## 478 1.62677199 4.399904
## 479 1.40132251 3.575396
## 480 3.26111441 4.780946
## 481 3.26492068 4.199313
## 482 1.65670958 3.895382
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## 484 4.34847396 5.778373
## 485 3.78072544 5.024230
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## 492 2.88892213 5.960457
## 493 1.54466337 3.713336
## 494 4.54505687 5.471034
## 495 3.31761614 6.151726
## 496 5.11664326 6.532764
## 497 2.59925843 5.811964
## 498 4.56641752 3.396775
## 499 6.11091611 7.417640
## 500 3.22665101 7.157814
```

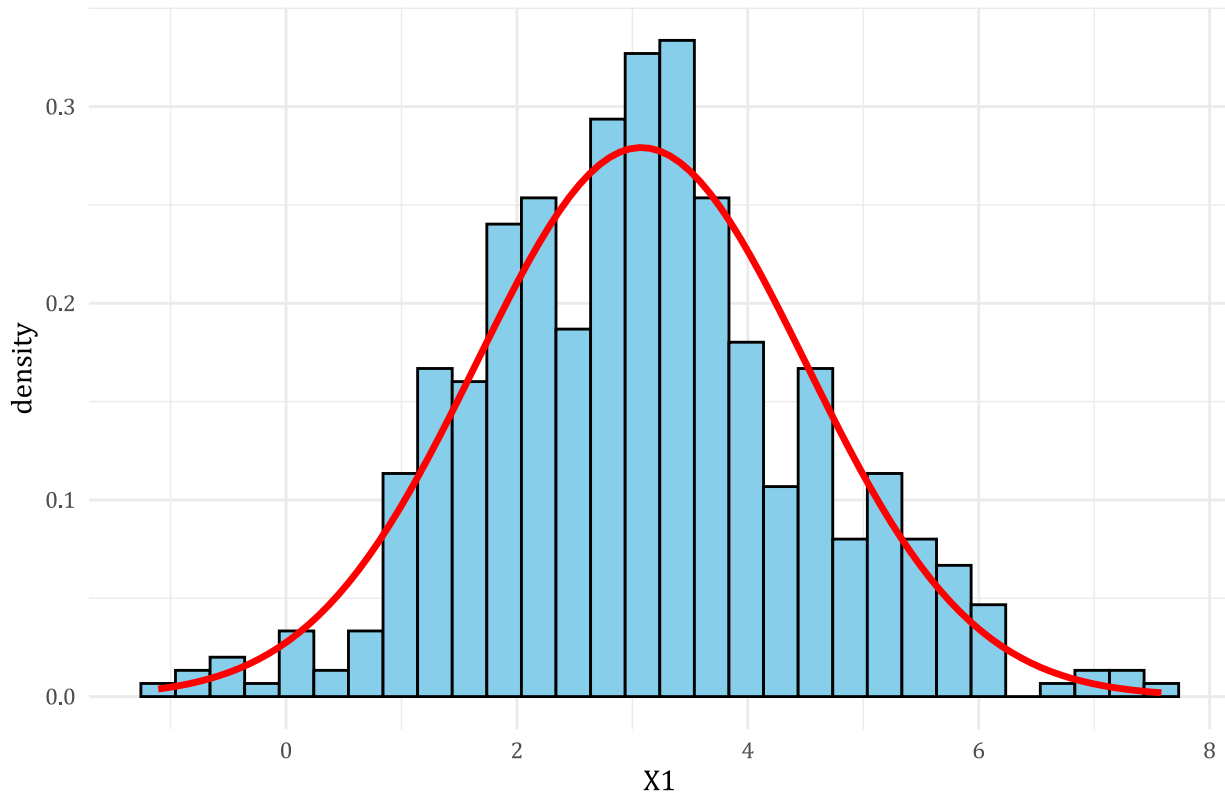
```
ggplot(samples_df, aes(x = X1, y = X2)) +
  geom_point(alpha = 0.6) +
  theme_minimal(base_family = "LM") +
  labs(title = "Biểu đồ tán xạ của dữ liệu được sinh", x = "X1", y = "X2")
```

Biểu đồ tán xạ của dữ liệu được sinh



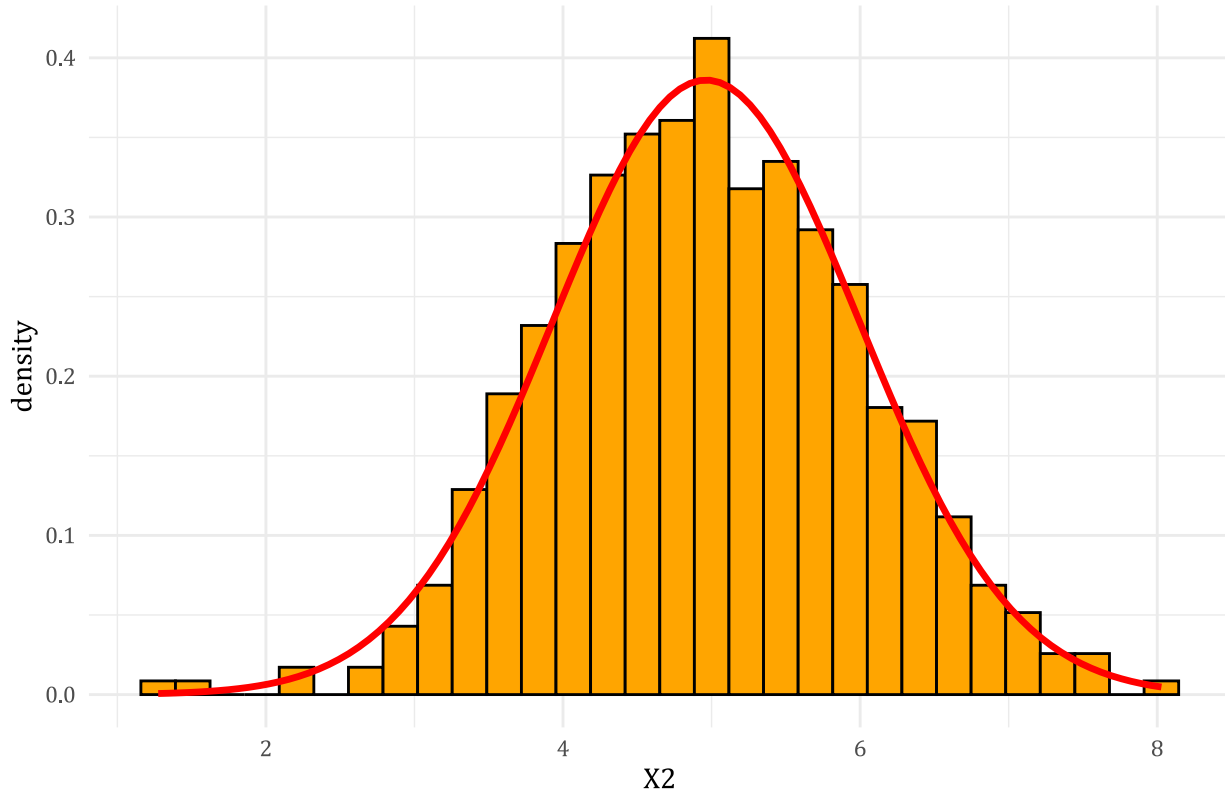
```
ggplot(samples_df, aes(x = X1)) +  
  geom_histogram(aes(y = after_stat(density)),  
    bins = 30,  
    fill = "skyblue",  
    color = "black"  
  ) +  
  stat_function(  
    fun = dnorm,  
    args = list(  
      mean = mean(samples_df$X1),  
      sd = sd(samples_df$X1)  
    ),  
    color = "red",  
    linewidth = 1.2  
  ) +  
  theme_minimal(base_family = "LM") +  
  labs(title = "Histogram của X1")
```

Histogram của X1



```
ggplot(samples_df, aes(x = X2)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 30,
    fill = "orange",
    color = "black"
  ) +
  stat_function(
    fun = dnorm,
    args = list(
      mean = mean(samples_df$X2),
      sd = sd(samples_df$X2)
    ),
    color = "red",
    linewidth = 1.2
  ) +
  theme_minimal(base_family = "LM") +
  labs(title = "Histogram của X2")
```

Histogram của X_2



- Hiệp phương sai $cov(X_1, X_2) = 0.5$ (Giá trị trên đường chéo phụ).

Hiệp phương sai dương nên hai biến biến thiên cùng chiều.

- Hệ số tương quan: $\rho = \frac{cov(X_1, X_2)}{\sqrt{DX_1 \cdot DX_2}} = \frac{0.5}{\sqrt{2 \cdot 1}} = \frac{1}{2\sqrt{2}} \approx 0.35$.

Hệ số tương quan ≈ 0.35 nên X_1 và X_2 tương quan không mạnh.

Câu 2. Sử dụng câu lệnh `set.seed(100)` và sinh mẫu ngẫu nhiên gồm 100 quan sát tuân theo phân phối chuẩn 3 chiều với giá trị trung bình $(0, 0, 1)^T$ và ma trận hiệp phương sai

$$\begin{pmatrix} 1 & .8 & .6 \\ .8 & 1 & .8 \\ .6 & .8 & 1 \end{pmatrix}$$

- (1) Đặt tên các biến lần lượt là A, B, C. Thêm một biến D vào mẫu trên, D gồm 100 quan sát được sinh ngẫu nhiên biết nó tuân theo phân phối chuẩn với hai tham số 0.5 và 0.5.

```
library(MASS)

set.seed(100)

n <- 100

mu <- c(0, 0, 1)

sigma <- matrix(
  c(
    1, 0.8, 0.6,
    0.8, 1, 0.8,
    0.6, 0.8, 1
  )
```

```

),
  byrow = TRUE, nrow = 3
)

samples <- mvrnorm(n = n, mu = mu, Sigma = sigma)

samples_df <- as.data.frame(samples)
colnames(samples_df) <- c("A", "B", "C")

samples_df$D <- rnorm(n = n, mean = 0.5, sd = sqrt(0.5))

samples_df

```

```

##           A           B           C           D
## 1    0.297846027  0.473292720  1.59562172 -0.811997902
## 2    0.438386912 -0.024318766  0.21918095 -0.220234017
## 3   -0.007978524 -0.167569018  1.41163961 -0.239005136
## 4   -0.324522357 -0.996377899 -0.07841324  0.039470084
## 5   -0.597059489 -0.402549035  1.70700974  0.770597499
## 6   -0.618210339 -0.014020609  0.73983416  1.229663283
## 7    0.078089841  0.717833071  1.77253857  0.592684003
## 8   -0.951693252 -0.396020550  0.37861559  0.766714833
## 9    0.812742204  1.652646081  0.70356174 -0.834801993
## 10   0.320518956  0.250110247  1.41661068  0.867850668
## 11   0.190518671 -0.440001593  1.03614635  1.107513027
## 12   0.133240582 -0.283773862  0.90551829 -0.067445170
## 13  -0.303618653  0.572642660  1.24603711  0.152131383
## 14  -1.422859578 -0.466479705  0.85546927  0.350832771
## 15  -0.534298112  0.392001491  0.76173532  0.172131126
## 16   0.274354666 -0.323186814  1.15952268  0.711764033
## 17   0.745949844  0.468085005  0.83531564  0.071865351
## 18  -0.714404399 -0.213708682  0.51418766  0.189571542
## 19   0.806109267  0.625013372  2.07671017 -0.009649779
## 20  -1.352068610 -2.141982586 -1.79673908  1.052112532
## 21  -0.195671196  0.852855945  1.49635907 -0.364285500
## 22  -0.487118452 -0.725504845  0.13364227  1.130169347
## 23  -0.487763531 -0.258682492  1.03453453  0.679550564
## 24  -0.470251908 -0.787962461  0.15852890  0.453460758
## 25   0.089920783  0.663592690  2.47204483  0.642457998
## 26   0.056864728  0.599187058  1.52084839  2.251998835
## 27   0.608129632  0.508621952  1.85838281  0.833579600
## 28   0.060595237  0.121496113  0.15949796  1.437763658
## 29   1.018765751  0.725536569  2.43876189  0.972770523
## 30  -0.177336359 -0.066015705  0.55621643  0.402847479
## 31  -0.037961055 -0.038808420  1.33572565 -0.497849076
## 32  -1.833096174 -1.859370159 -0.07249249  1.348784891
## 33   0.383232233  0.217515038  0.76693058  0.711488643
## 34  -0.356248436  0.563133776  1.05536355  0.822601122
## 35   1.158053135  0.758178685  0.95220037  0.809768239
## 36   0.303306890  0.378394870  0.90701180  0.347957095
## 37  -0.289191258 -0.063959876  0.84580123  0.279296081
## 38  -0.032664649 -0.641164228  0.55988007  0.010236334
## 39  -0.850169911 -1.216454836  0.18574553 -1.194390258
## 40  -0.613740592 -0.997701087 -0.02170766 -0.620251549
## 41  -0.203410342  0.300108400  1.16189611  1.743622802
## 42  -1.785947360 -2.090343201  1.12499084 -1.170813271

```

## 43	1.382793972	2.166951939	2.24252819	0.778014635
## 44	-0.502908302	-0.138513937	-0.09327399	0.900178502
## 45	0.073782146	0.403511017	1.95196154	-0.238212670
## 46	-1.239409819	-2.045414873	0.75666339	1.223450347
## 47	0.073777598	0.441670404	1.46495325	0.137491799
## 48	-1.587779004	-0.905741055	-0.12615507	0.533754926
## 49	0.081870541	0.081093932	0.70710913	1.484912770
## 50	2.110378001	1.073558590	2.99038573	0.194176935
## 51	0.736814249	0.648529139	0.81136255	0.404421906
## 52	2.192201175	1.811084132	1.71326293	1.417637325
## 53	-0.207484928	-0.094411230	0.80856719	0.449988824
## 54	-1.566131891	-2.032031319	-0.54448378	0.202429530
## 55	2.216922268	1.986824385	2.99309464	1.307049506
## 56	-1.085075887	-0.809858684	0.21645746	0.197975967
## 57	0.882304429	1.890619208	1.98369106	1.486887443
## 58	-1.359625940	-2.285882071	-0.27118237	0.290564044
## 59	-1.292567371	-0.967438211	-0.52877381	0.912002562
## 60	0.725128512	0.943584690	1.60081772	-1.392379922
## 61	0.155861502	0.189671758	1.37255393	-0.490429626
## 62	0.152546975	-0.054420540	1.09974770	2.836387657
## 63	0.064960767	0.711469215	1.22351755	1.105833209
## 64	-1.870503639	-2.563751900	-1.58128728	1.320962601
## 65	0.026213638	-0.377299078	1.02018718	0.697238163
## 66	0.420109503	0.547964677	1.98341107	0.490419767
## 67	-0.539914548	-0.859606482	0.68591918	-1.392485660
## 68	-0.624004479	0.059564302	0.99352788	0.585782273
## 69	0.132201254	0.103710577	0.95104205	0.146062479
## 70	0.630146928	-0.224126861	0.87312036	1.475579607
## 71	-0.032358769	-0.578018787	0.40229700	0.212924081
## 72	1.089394127	0.906599607	1.90924439	1.487072944
## 73	1.148733041	0.809866832	2.23018910	1.022266350
## 74	-1.392420699	-1.451804189	-0.65134317	1.020864863
## 75	1.889127731	1.797832948	2.93803943	-0.018371463
## 76	-1.051445463	0.221348886	1.77484057	-0.842368720
## 77	1.077335717	1.117513265	1.75680052	0.815284278
## 78	-0.380952161	-0.303391664	0.95231217	0.822625150
## 79	-0.232887021	-1.128701642	-0.36729693	-0.579184395
## 80	2.384272293	2.747020237	1.44461720	0.443019485
## 81	-1.168023614	-1.183753747	0.94081821	0.259576195
## 82	0.373292123	-0.072389930	0.84569288	0.537958057
## 83	1.450043459	0.904351858	2.33911723	0.010940308
## 84	1.638208460	1.536137405	3.10662694	0.285396701
## 85	-0.247256495	-0.864668455	0.19799471	-0.345002537
## 86	0.483075002	0.101245559	0.84962236	1.221062181
## 87	-0.130846862	-0.393802921	0.95251228	0.161688846
## 88	-0.862162107	-0.764723669	0.40186985	0.033668700
## 89	-1.906818468	-1.839551591	0.06436420	0.980040406
## 90	-0.131485250	-0.004453961	1.42736633	0.425502534
## 91	0.517434155	0.293388886	1.72584154	0.179173157
## 92	-1.475284675	-1.175999841	-0.25094725	1.448540353
## 93	1.152422130	0.967513449	1.29923220	1.197177023
## 94	0.963554708	0.993972178	2.20144775	-0.345321249
## 95	1.142942273	0.823991067	0.44775593	0.093513605
## 96	-2.406088842	-2.234741380	-1.02149335	1.194372751
## 97	1.483547482	0.714041616	1.07433541	-0.671946917
## 98	-0.013573956	-0.907801717	0.84158143	-0.291709938

```
## 99    0.679153790  0.897515637  2.65001238 -0.327897877
## 100   1.087327447  1.501789480  1.57129904 -0.305431118
```

(2) Tính giá trị trung bình mẫu, ma trận tương quan mẫu và ma trận hiệp phương sai Mẫu

Giá trị trung bình mẫu

```
samples_mean <- colMeans(samples_df)
samples_mean
```

```
##           A           B           C           D
## 0.004402080 -0.006440735  0.994437413  0.441136611
```

Ma trận tương quan mẫu

```
samples_cor <- cor(samples_df)
samples_cor
```

```
##           A           B           C           D
## A  1.00000000  0.879170642  0.72872852 -0.024679815
## B  0.87917064  1.000000000  0.78095664 -0.004130266
## C  0.72872852  0.780956636  1.00000000 -0.086675328
## D -0.02467981 -0.004130266 -0.08667533  1.000000000
```

Ma trận hiệp phương sai mẫu

```
samples_cov <- cov(samples_df)
samples_cov
```

```
##           A           B           C           D
## A  0.98614256  0.918540588  0.67664631 -0.018731241
## B  0.91854059  1.106905752  0.76826021 -0.003321147
## C  0.67664631  0.768260211  0.87428333 -0.061940740
## D -0.01873124 -0.003321147 -0.06194074  0.584130109
```

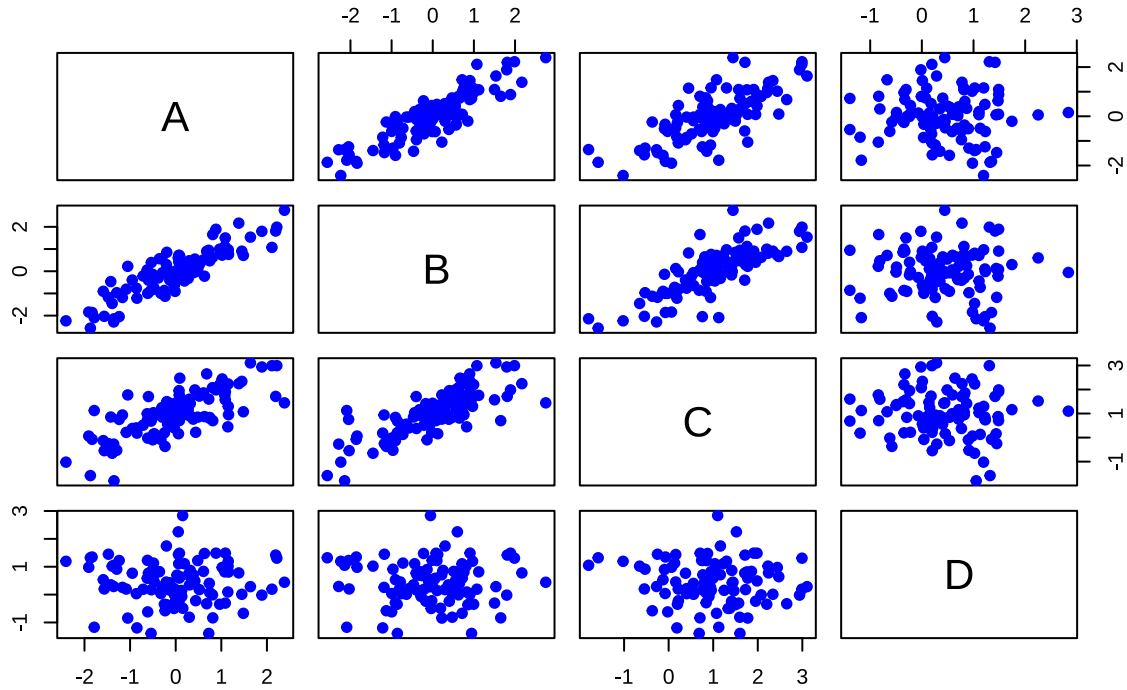
(3) Vẽ biểu đồ phân tán của các biến và biểu đồ nhiệt của ma trận tương quan mẫu.

```
library(ggplot2)
library(showtext)

font_add("LM", "C:/Windows/Fonts/cambria.ttc")
showtext_auto()

pairs(samples_df,
  main = "Biểu đồ phân tán của các biến A, B, C, D",
  pch = 19,
  col = "blue"
)
```

Biểu đồ phân tán của các biến A, B, C, D



```
cor_mat <- as.data.frame(as.table(samples_cor))
colnames(cor_mat) <- c("Z1", "Z2", "Correlation")

ggplot(cor_mat, aes(x = Z1, y = Z2, fill = Correlation)) +
  geom_tile() +
  geom_text(aes(label = round(Correlation, 4), color = "black")) +
  scale_y_discrete(limits = rev(levels(cor_mat$Z2))) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    limits = c(-1, 1),
    name = "Hệ số tương quan"
  ) +
  theme_minimal(base_family = "LM") +
  labs(title = "Biểu đồ nhiệt của ma trận tương quan mẫu")
```

Biểu đồ nhiệt của ma trận tương quan mẫu

